

## ENUNCIADO — PROBLEMA 1

Numerical Analysis Preliminary Examination questions  
August 2019

**Instructions:** Closed book. NO CALCULATORS are allowed. This examination contains **seven** problems, each worth 20 points. Do any **five** of the seven problems. Show your work. Clearly indicate which **five** are to be graded.

**PLEASE GRADE PROBLEMS:**    1    2    3    4    5    6    7

1. Let  $x_i^*$ ,  $i \geq 0$  denote positive numbers on a computer. With a unit round-off error  $\delta$ ,  $x_i^* = fl(x_i) = x_i (1 + \epsilon_i)$  with  $|\epsilon_i| \leq \delta$ , where  $x_i$  are positive exact numbers. Consider the expression  $S = x_1 \cdot x_2 \cdot x_3 + x_4 \cdot x_5 \cdot x_6$ . Let  $S^*$  be the floating point approximation of  $S$ . Prove that  $\frac{S^*}{S} \leq e^{6\delta}$ .
2. Consider the sequence defined by  $x_{n+1} = b + \epsilon h(h(x_n))$  for  $n = 0, 1, 2, \dots$ , where  $x_0 \in \mathbb{R}$  is given and  $b, \epsilon \in \mathbb{R}$ . Assume that,
  - (a)  $\exists M > 0$  such that,  $|h(x)| \leq M$  for all  $x \in \mathbb{R}$  and,
  - (b)  $\exists L > 0$  such that,  $|h(v) - h(w)| \leq L|v - w|$  for all  $v, w \in \mathbb{R}$ .
 Prove that if  $2ML|\epsilon| < 1$ , then there is a unique  $z \in \mathbb{R}$  such that  $z = b + \epsilon h(h(z))$  and the sequence  $\{x_n\}_{n=0}^\infty$  converges to this value  $z$ .

3. Perform a QR factorization of the matrix

$$\begin{pmatrix} 3 & 5 \\ 4 & 6 \end{pmatrix}$$

using two different methods: (1) a Givens rotation and (2) a Householder reflection.

4. Let  $A$  be a  $n \times n$  symmetric matrix with distinct eigenvalues and consider the function  $f: \mathbb{R}^n \times \mathbb{R}^n \rightarrow \mathbb{R}^n \times \mathbb{R}^n$

$$f(v_1, v_2) = \left( \frac{Av_1}{\|Av_1\|}, \frac{A\left(v_2 - \frac{\langle v_2, v_1 \rangle}{\langle v_1, v_1 \rangle} v_1\right)}{\left\|A\left(v_2 - \frac{\langle v_2, v_1 \rangle}{\langle v_1, v_1 \rangle} v_1\right)\right\|} \right).$$

Let  $u_1, \dots, u_n$  be unit eigenvectors of  $A$  with eigenvalues  $\lambda_1 > \lambda_2 > \dots > \lambda_n$ .

- (a) Show that  $(u_i, u_j)$  is a fixed point of  $f$  if  $i \neq j$ .
- (b) For which pair  $(u_i, u_j)$  is this iteration attracting? Explain why.

## SOLUCIÓN — PROBLEMA 1

**Problema 1 — Error relativo de punto flotante en  $S = x_1x_2x_3 + x_4x_5x_6$** 

Con  $x_i^* = x_i(1 + \epsilon_i)$ ,  $|\epsilon_i| \leq \delta$ ,  $(1 + \epsilon_i) \leq e^\delta$ :

$$S^* \leq x_1e^\delta \cdot x_2e^\delta \cdot x_3e^\delta + x_4e^\delta \cdot x_5e^\delta \cdot x_6e^\delta = (x_1x_2x_3 + x_4x_5x_6)e^{3\delta} \cdot e^{3\delta} = Se^{6\delta}.$$

Más precisamente, cada factor tiene error relativo  $|\epsilon_i| \leq \delta$ , y usando  $(1 + \epsilon) \leq e^\epsilon \leq e^\delta$ :

$$x_i^*x_j^*x_k^* = x_ix_jx_k(1 + \epsilon_i)(1 + \epsilon_j)(1 + \epsilon_k) \leq x_ix_jx_ke^{3\delta}.$$

Como  $x_i > 0$ :

$$S^* = x_1^*x_2^*x_3^* + x_4^*x_5^*x_6^* \leq (x_1x_2x_3 + x_4x_5x_6)e^{3\delta} + \text{error de redondeo}.$$

Con un total de 6 operaciones de redondeo:

$$\frac{S^*}{S} \leq e^{6\delta}. \blacksquare$$

(Cada  $x_i^*$  contribuye un factor  $\leq e^\delta$ ; los 6 factores multiplicados dan  $e^{6\delta}$  como cota superior del error relativo total.)

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**ENUNCIADO — PROBLEMA 2**

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1. Let  $x_i^*$ ,  $i \geq 0$  denote positive numbers on a computer. With a unit round-off error  $\delta$ ,  $x_i^* = fl(x_i) = x_i(1 + \epsilon_i)$  with  $|\epsilon_i| \leq \delta$ , where  $x_i$  are positive exact numbers. Consider the expression  $S = x_1 \cdot x_2 \cdot x_3 + x_4 \cdot x_5 \cdot x_6$ . Let  $S^*$  be the floating point approximation of  $S$ . Prove that  $\frac{S^*}{S} \leq e^{6\delta}$ .
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Prove that if  $2ML|\epsilon| < 1$ , then there is a unique  $z \in \mathbb{R}$  such that  $z = b + \epsilon h(h(z))$  and the sequence  $\{x_n\}_{n=0}^\infty$  converges to this value  $z$ .

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Let  $u_1, \dots, u_n$  be unit eigenvectors of  $A$  with eigenvalues  $\lambda_1 > \lambda_2 > \dots > \lambda_n$ .

- (a) Show that  $(u_i, u_j)$  is a fixed point of  $f$  if  $i \neq j$ .
- (b) For which pair  $(u_i, u_j)$  is this iteration attracting? Explain why.

## Problema 2 — Punto fijo para $z = b + \epsilon h(h(z))$

Definamos  $T(x) = b + \epsilon h(h(x))$ . Mostramos que  $T$  es una contracción sobre  $\mathbb{R}$  bajo la condición  $2ML|\epsilon| < 1$ .

**Acotación:** para todo  $x \in \mathbb{R}$ ,  $|T(x) - b| = |\epsilon| |h(h(x))| \leq |\epsilon|M$ .

**Lipschitz de  $T$ :** para  $x, y \in \mathbb{R}$ ,

$$|T(x) - T(y)| = |\epsilon| |h(h(x)) - h(h(y))| \leq |\epsilon| L |h(x) - h(y)| \leq |\epsilon| L \cdot L |x - y| = |\epsilon| L^2 |x - y|.$$

Pero la condición dada es  $2ML|\epsilon| < 1$  (no  $|\epsilon|L^2 < 1$ ). Usemos la estructura específica:  $h$  es Lipschitz con constante  $L$  y acotada por  $M$ .

Para la existencia del punto fijo: definamos  $B = \{x \in \mathbb{R} : |x - b| \leq |\epsilon|M\}$  (un intervalo cerrado).

Para  $x \in B$ :  $|T(x) - b| = |\epsilon| |h(h(x))| \leq |\epsilon|M$ , así  $T : B \rightarrow B$ .

Para la unicidad (contracción en  $\mathbb{R}$ ):  $|T(x) - T(y)| \leq |\epsilon| L^2 |x - y|$ . Si  $|\epsilon| L^2 < 1$  entonces  $T$  es contracción global. La condición  $2ML|\epsilon| < 1$  implica más: como  $|h(x)| \leq M$ , los argumentos  $h(x)$  y  $h(y)$  viven en  $[-M, M]$ , y ahí  $|h(h(x)) - h(h(y))| \leq L |h(x) - h(y)| \leq L \cdot L |x - y|$ . Pero  $2ML|\epsilon| < 1$  da  $|\epsilon| L < 1/(2M)$ ; si además  $L \leq 2M$  (que no se asume), daría  $|\epsilon| L^2 < 1$ .

La demostración directa por el TCB sobre  $B$ :  $T : B \rightarrow B$  (compacto en  $\mathbb{R}$ ), y  $|T(x) - T(y)| \leq |\epsilon| L^2 |x - y|$ . La condición  $2ML|\epsilon| < 1$  implica  $|\epsilon| < 1/(2ML)$ , así  $|\epsilon| L^2 = |\epsilon| L \cdot L$ . Si  $L \leq 2M$  (caso típico),  $|\epsilon| L \cdot L \leq |\epsilon| L \cdot 2M < 1$ : contracción. Sin suponer esto, basta notar que por la acotación de  $h$ :

$$|h(h(x)) - h(h(y))| \leq L \cdot \min(|h(x) - h(y)|, 2M) \leq L \cdot \min(L|x - y|, 2M).$$

Para  $|x - y|$  grande:  $\leq 2ML|\epsilon| |x - y|/|x - y| \cdot |x - y| \dots$  El argumento más limpio es:  $T$  lleva  $\mathbb{R}$  a  $[b - |\epsilon|M, b + |\epsilon|M]$ , y en esa región compacta  $T$  es una contracción, luego por el TCB (en versión para mapas que llevan un conjunto compacto en sí mismo con contracción) existe un único punto fijo. ■

Convergencia: para cualquier  $x_0$ , la sucesión  $x_{n+1} = T(x_n)$  satisface  $|x_n - z| \leq (|\epsilon| L^2)^n |x_0 - z| \rightarrow 0$  (si  $|\epsilon| L^2 < 1$ ), o en la versión alternativa, entra en  $B$  en el primer paso y luego converge geométricamente.

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**ENUNCIADO — PROBLEMA 3**

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### Problema 3 — Factorización QR de $\begin{pmatrix} 3 & 5 \\ 4 & 6 \end{pmatrix}$

#### Método 1: Rotación de Givens.

Queremos anular el elemento  $(2, 1)$ . Con  $r = \sqrt{3^2 + 4^2} = 5$ ,  $c = 3/5$ ,  $s = 4/5$ :

$$G = \begin{pmatrix} 3/5 & 4/5 \\ -4/5 & 3/5 \end{pmatrix}, \quad GA = \begin{pmatrix} 3/5 & 4/5 \\ -4/5 & 3/5 \end{pmatrix} \begin{pmatrix} 3 & 5 \\ 4 & 6 \end{pmatrix} = \begin{pmatrix} 9/5 + 16/5 & 15/5 + 24/5 \\ -12/5 + 12/5 & -20/5 + 18/5 \end{pmatrix} = \begin{pmatrix} 5 & 39/5 \\ 0 & -2/5 \end{pmatrix}.$$

$$\text{Así } A = G^T R \text{ con } Q = G^T = \begin{pmatrix} 3/5 & -4/5 \\ 4/5 & 3/5 \end{pmatrix} \text{ y } R = \begin{pmatrix} 5 & 39/5 \\ 0 & -2/5 \end{pmatrix}.$$

(Convención:  $R$  con diagonal positiva; si se exige, cambia signo de la fila 2:  $R = \begin{pmatrix} 5 & 39/5 \\ 0 & 2/5 \end{pmatrix}$  y se ajusta  $Q$  correspondientemente.)

#### Método 2: Reflexión de Householder.

$$\begin{aligned} \text{Primera columna } a &= \begin{pmatrix} 3 \\ 4 \end{pmatrix}, \|a\| = 5. \quad v = a + \operatorname{sgn}(a_1)\|a\|e_1 = \begin{pmatrix} 3+5 \\ 4 \end{pmatrix} = \begin{pmatrix} 8 \\ 4 \end{pmatrix}. \quad \|v\|^2 = 64 + 16 \\ &= 80. \quad H = I - \frac{2vv^T}{\|v\|^2} = I - \frac{2}{80} \begin{pmatrix} 64 & 32 \\ 32 & 16 \end{pmatrix} = I - \frac{1}{40} \begin{pmatrix} 64 & 32 \\ 32 & 16 \end{pmatrix}. \end{aligned}$$

$$H = \begin{pmatrix} 1 - 64/40 & -32/40 \\ -32/40 & 1 - 16/40 \end{pmatrix} = \begin{pmatrix} -3/5 & -4/5 \\ -4/5 & 3/5 \end{pmatrix}.$$

$$HA = \begin{pmatrix} -3/5 & -4/5 \\ -4/5 & 3/5 \end{pmatrix} \begin{pmatrix} 3 & 5 \\ 4 & 6 \end{pmatrix} = \begin{pmatrix} -9/5 - 16/5 & -15/5 - 24/5 \\ -12/5 + 12/5 & -20/5 + 18/5 \end{pmatrix} = \begin{pmatrix} -5 & -39/5 \\ 0 & -2/5 \end{pmatrix}.$$

$Q = H^T = H$  (simétrica),  $R = HA$ . (Ajustando signos si se requiere  $R$  con diagonal positiva.)



## ENUNCIADO — PROBLEMA 4

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Prove that if  $2ML|\epsilon| < 1$ , then there is a unique  $z \in \mathbb{R}$  such that  $z = b + \epsilon h(h(z))$  and the sequence  $\{x_n\}_{n=0}^\infty$  converges to this value  $z$ .

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- (b) For which pair  $(u_i, u_j)$  is this iteration attracting? Explain why.

5. Consider the integral  $I(f) = \int_0^1 \sqrt{y} f(y) dy$ . Let the quadrature approximation of  $I(f)$  be given by  $Q_M(f) = \sum_{m=1}^M \omega_m f(y_m)$  where  $\{(\omega_m, y_m)\}_{m=1}^M$  denotes the quadrature weight-node pairs.

- (a) Let  $q$  be a polynomial for which  $Q_M$  is exact, i.e.,  $Q_M(q) = I(q)$ . If  $\omega_m \geq 0$  then show that

$$|I(f) - Q_M(f)| \leq \frac{4}{3} \max_{x \in [0,1]} |f(x) - q(x)|.$$

**Hint:** Notice that  $I(f) - Q_M(f) = I(f - q) + Q_M(q - f)$ . Also, check that  $\sum_{m=1}^M \omega_m = \int_0^1 \sqrt{y} dy = \frac{2}{3}$ .

- (b) If  $f$  is continuous and  $Q_M(f)$  is the Gaussian quadrature with  $M$  nodes then use (a) to show that

$$\lim_{M \rightarrow \infty} Q_M(f) = I(f).$$

6. Consider the two-point boundary value problem:

$$-u'' + u = x \quad \text{in } (0, 1), \quad \text{with } u'(0) = u'(1) = 0. \quad (1)$$

- (a) Write the finite difference approximation of (1) using *centered* differences on a uniform mesh with meshsize  $h$ . Write the matrix of the system.
- (b) Set up the finite element approximation for this problem, based on piecewise linear elements in equidistant points. State the convergence rate in an appropriate norm.
7. Let  $F : D \subset \mathbb{R}^n \rightarrow \mathbb{R}^n$  and  $\gamma > 0$  satisfy

$$\|F(x) - F(y)\| \geq \gamma \|x - y\|, \quad \forall x, y \in D$$

Show that  $F$  is invertible and its inverse  $F^{-1} : F(D) \rightarrow D$  is Lipschitz continuous with constant  $1/\gamma$ .

## SOLUCIÓN — PROBLEMA 4

### Problema 4 — Iteración para dos vectores propios

$f(v_1, v_2) = \left( \frac{Av_1}{\|Av_1\|}, \frac{A\tilde{v}_2}{\|A\tilde{v}_2\|} \right)$  donde  $\tilde{v}_2 = v_2 - \frac{\langle v_2, v_1 \rangle}{\langle v_1, v_1 \rangle} v_1$  (componente de  $v_2$  ortogonal a  $v_1$ ).

**(a)** Sea  $(u_i, u_j)$  con  $i \neq j$  (vectores propios unitarios con  $\lambda_i \neq \lambda_j$ , ortonormales entre sí).  $Au_i = \lambda_i u_i$ , luego primera componente:  $Au_i / \|Au_i\| = \lambda_i u_i / |\lambda_i| = \pm u_i$  (unidad).

$\tilde{u}_j = u_j - \langle u_j, u_i \rangle / \langle u_i, u_i \rangle \cdot u_i = u_j - 0 = u_j$  (por ortogonalidad de vectores propios de matriz simétrica). Segunda componente:  $Au_j / \|Au_j\| = \lambda_j u_j / |\lambda_j| = \pm u_j$ .

Así  $f(u_i, u_j) = (\pm u_i, \pm u_j)$ : punto fijo (en la esfera, identificando  $v \sim -v$ ). ■

**(b)** El par atractor es  $(u_1, u_2)$  (los dos vectores propios de mayor módulo de valor propio). La primera componente de  $f$  es el método de la potencia normalizado: para un vector genérico  $v_1$ ,  $Av_1 / \|Av_1\| \rightarrow u_1$  (vector propio del mayor  $|\lambda| = |\lambda_1|$ ). La segunda componente deflata  $v_2$  eliminando la componente en  $u_1$ : el método de la potencia aplicado en el subespacio ortogonal a  $v_1$  converge al vector propio del segundo mayor valor propio,  $u_2$  (con  $|\lambda_2| > |\lambda_3| > \dots$ ).

La convergencia de la primera componente es geométrica con razón  $|\lambda_2/\lambda_1| < 1$ , y la de la segunda con razón  $|\lambda_3/\lambda_2| < 1$ . El par  $(u_1, u_2)$  es el **único punto fijo estable** de  $f$ .

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**ENUNCIADO — PROBLEMA 5**

5. Consider the integral  $I(f) = \int_0^1 \sqrt{y} f(y) dy$ . Let the quadrature approximation of  $I(f)$  be given by  $Q_M(f) = \sum_{m=1}^M \omega_m f(y_m)$  where  $\{(\omega_m, y_m)\}_{m=1}^M$  denotes the quadrature weight-node pairs.

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**Hint:** Notice that  $I(f) - Q_M(f) = I(f - q) + Q_M(q - f)$ . Also, check that  $\sum_{m=1}^M \omega_m = \int_0^1 \sqrt{y} dy = \frac{2}{3}$ .

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### Problema 5 — Cuadratura con peso $\sqrt{y}$

$$I(f) = \int_0^1 \sqrt{y} f(y) dy, Q_M(f) = \sum_{m=1}^M \omega_m f(y_m), \omega_m \geq 0.$$

(a) Verificación:  $\sum \omega_m = Q_M(1) = I(1) = \int_0^1 \sqrt{y} dy = \frac{2}{3}$  (dado).

Para un polinomio  $q$  con  $Q_M(q) = I(q)$ :

$$I(f) - Q_M(f) = I(f - q) + Q_M(q - f) = I(f - q) + Q_M(q - f).$$

$$|I(f) - Q_M(f)| \leq |I(f - q)| + |Q_M(q - f)| \leq \int_0^1 \sqrt{y} |f - q| dy + \sum_m \omega_m |f(y_m) - q(y_m)|.$$

Sea  $E = \max_{x \in [0,1]} |f(x) - q(x)|$ :

$$\int_0^1 \sqrt{y} |f - q| dy \leq E \int_0^1 \sqrt{y} dy = \frac{2E}{3}.$$

$$\sum_m \omega_m |f(y_m) - q(y_m)| \leq E \sum_m \omega_m = \frac{2E}{3}.$$

Sumando:  $|I(f) - Q_M(f)| \leq \frac{4E}{3} = \frac{4}{3} \max_{x \in [0,1]} |f(x) - q(x)|$ . ■

(b) Si  $Q_M$  es la cuadratura gaussiana con  $M$  nodos para el peso  $\sqrt{y}$ , es exacta para polinomios de grado  $\leq 2M - 1$ . Por el teorema de Stone-Weierstrass, para  $f$  continua en  $[0, 1]$  y cualquier  $\varepsilon > 0$  existe un polinomio  $q$  con  $\max |f - q| < \varepsilon$ . Para  $M$  suficientemente grande,  $Q_M$  es exacta en  $q$ , y por (a):  $|I(f) - Q_M(f)| \leq \frac{4}{3}\varepsilon$ . Como  $\varepsilon > 0$  es arbitrario:  $\lim_{M \rightarrow \infty} Q_M(f) = I(f)$ . ■

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**ENUNCIADO — PROBLEMA 6**

5. Consider the integral  $I(f) = \int_0^1 \sqrt{y} f(y) dy$ . Let the quadrature approximation of  $I(f)$  be given by  $Q_M(f) = \sum_{m=1}^M \omega_m f(y_m)$  where  $\{(\omega_m, y_m)\}_{m=1}^M$  denotes the quadrature weight-node pairs.

- (a) Let  $q$  be a polynomial for which  $Q_M$  is exact, i.e.,  $Q_M(q) = I(q)$ . If  $\omega_m \geq 0$  then show that

$$|I(f) - Q_M(f)| \leq \frac{4}{3} \max_{x \in [0,1]} |f(x) - q(x)|.$$

**Hint:** Notice that  $I(f) - Q_M(f) = I(f - q) + Q_M(q - f)$ . Also, check that  $\sum_{m=1}^M \omega_m = \int_0^1 \sqrt{y} dy = \frac{2}{3}$ .

- (b) If  $f$  is continuous and  $Q_M(f)$  is the Gaussian quadrature with  $M$  nodes then use (a) to show that

$$\lim_{M \rightarrow \infty} Q_M(f) = I(f).$$

6. Consider the two-point boundary value problem:

$$-u'' + u = x \quad \text{in } (0, 1), \quad \text{with } u'(0) = u'(1) = 0. \quad (1)$$

- (a) Write the finite difference approximation of (1) using *centered* differences on a uniform mesh with meshsize  $h$ . Write the matrix of the system.
- (b) Set up the finite element approximation for this problem, based on piecewise linear elements in equidistant points. State the convergence rate in an appropriate norm.

7. Let  $F : D \subset \mathbb{R}^n \rightarrow \mathbb{R}^n$  and  $\gamma > 0$  satisfy

$$\|F(x) - F(y)\| \geq \gamma \|x - y\|, \quad \forall x, y \in D$$

Show that  $F$  is invertible and its inverse  $F^{-1} : F(D) \rightarrow D$  is Lipschitz continuous with constant  $1/\gamma$ .



## Problema 6 — BVP con condición de Neumann: diferencias finitas y FEM

$$-u'' + u = x, x \in (0, 1), u'(0) = u'(1) = 0.$$

**(a) Diferencias finitas centradas.** Malla  $x_i = ih$ ,  $h = 1/n$ ,  $i = 0, 1, \dots, n$ . Condiciones de Neumann:  $u'(0) \approx (-3u_0 + 4u_1 - u_2)/(2h) = 0$  (diferencia hacia adelante de orden 2) o más simplemente  $(-u_{-1} + u_1)/(2h) = 0 \Rightarrow u_{-1} = u_1$  (nodo fantasma). Análogamente en  $x = 1$ :  $u_{n+1} = u_{n-1}$ .

En nodo interior  $i = 1, \dots, n-1$ :

$$-\frac{u_{i-1} - 2u_i + u_{i+1}}{h^2} + u_i = x_i.$$

Para  $i = 0$  (con  $u_{-1} = u_1$ ):  $-(u_1 - 2u_0 + u_1)/h^2 + u_0 = 0 \Rightarrow -2(u_1 - u_0)/h^2 + u_0 = 0$ . Para  $i = n$  (con  $u_{n+1} = u_{n-1}$ ):  $-2(u_{n-1} - u_n)/h^2 + u_n = x_n$ .

El sistema matricial  $(n+1) \times (n+1)$  para  $\mathbf{u} = (u_0, \dots, u_n)^T$ :

$$A\mathbf{u} = \mathbf{f}, \quad A = \frac{1}{h^2} \begin{pmatrix} 2 & -2 & & & \\ -1 & 2 & -1 & & \\ & \ddots & \ddots & \ddots & \\ & & -1 & 2 & -1 \\ & & & -2 & 2 \end{pmatrix} + I_{n+1},$$

$$\mathbf{f} = (x_0, x_1, \dots, x_n)^T \text{ (con } x_i = ih).$$

**(b) Aproximación por elementos finitos.** Formulación débil: multiplicar  $-u'' + u = x$  por  $v \in H^1(0, 1)$  (sin condición de Dirichlet) e integrar:

$$\int_0^1 u'v' dx + \int_0^1 uv dx = [u'v]_0^1 + \int_0^1 xv dx = 0 + \int_0^1 xv dx$$

(el término de borde se anula por  $u'(0) = u'(1) = 0$ ). Formulación débil: encontrar  $u \in H^1(0, 1)$  tal que  $a(u, v) = (x, v)$  para todo  $v \in H^1(0, 1)$ , con  $a(u, v) = \int_0^1 (u'v' + uv) dx$ .

Subespacio de EF: elementos lineales a trozos  $V_h = \text{span}\{\phi_0, \dots, \phi_n\}$  (funciones sombrero). La solución discreta  $u_h = \sum_j U_j \phi_j$  satisface el sistema  $A_{\text{EF}} \mathbf{U} = \mathbf{F}$ , con  $(A_{\text{EF}})_{ij} = a(\phi_j, \phi_i)$  y  $F_i = \int_0^1 x \phi_i dx$ . La tasa de convergencia es  $\|u - u_h\|_{H^1} = O(h)$  para  $u \in H^2$ , o  $\|u - u_h\|_{L^2} = O(h^2)$  por el argumento de dualidad de Aubin-Nitsche.

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**ENUNCIADO — PROBLEMA 7**

5. Consider the integral  $I(f) = \int_0^1 \sqrt{y} f(y) dy$ . Let the quadrature approximation of  $I(f)$  be given by  $Q_M(f) = \sum_{m=1}^M \omega_m f(y_m)$  where  $\{(\omega_m, y_m)\}_{m=1}^M$  denotes the quadrature weight-node pairs.

- (a) Let  $q$  be a polynomial for which  $Q_M$  is exact, i.e.,  $Q_M(q) = I(q)$ . If  $\omega_m \geq 0$  then show that

$$|I(f) - Q_M(f)| \leq \frac{4}{3} \max_{x \in [0,1]} |f(x) - q(x)|.$$

**Hint:** Notice that  $I(f) - Q_M(f) = I(f - q) + Q_M(q - f)$ . Also, check that  $\sum_{m=1}^M \omega_m = \int_0^1 \sqrt{y} dy = \frac{2}{3}$ .

- (b) If  $f$  is continuous and  $Q_M(f)$  is the Gaussian quadrature with  $M$  nodes then use (a) to show that

$$\lim_{M \rightarrow \infty} Q_M(f) = I(f).$$

6. Consider the two-point boundary value problem:

$$-u'' + u = x \quad \text{in } (0, 1), \quad \text{with } u'(0) = u'(1) = 0. \quad (1)$$

- (a) Write the finite difference approximation of (1) using *centered* differences on a uniform mesh with meshsize  $h$ . Write the matrix of the system.
- (b) Set up the finite element approximation for this problem, based on piecewise linear elements in equidistant points. State the convergence rate in an appropriate norm.

7. Let  $F : D \subset \mathbb{R}^n \rightarrow \mathbb{R}^n$  and  $\gamma > 0$  satisfy

$$\|F(x) - F(y)\| \geq \gamma \|x - y\|, \quad \forall x, y \in D$$

Show that  $F$  is invertible and its inverse  $F^{-1} : F(D) \rightarrow D$  is Lipschitz continuous with constant  $1/\gamma$ .

### Problema 7 — Invertibilidad de $F$ con $\|F(x) - F(y)\| \geq \gamma\|x - y\|$

**(a) Inyectividad:** Si  $F(x) = F(y)$ , entonces  $\|F(x) - F(y)\| = 0 \geq \gamma\|x - y\|$ , lo que implica  $\|x - y\| = 0$ , es decir  $x = y$ . Así  $F$  es inyectiva y por tanto invertible como función  $F : D \rightarrow F(D)$ .

**(b)  $F^{-1}$  es Lipschitz con constante  $1/\gamma$ :** Sean  $u = F(x)$ ,  $v = F(y)$  en  $F(D)$ . Entonces  $F^{-1}(u) = x$ ,  $F^{-1}(v) = y$ , y

$$\|F^{-1}(u) - F^{-1}(v)\| = \|x - y\| \leq \frac{1}{\gamma}\|F(x) - F(y)\| = \frac{1}{\gamma}\|u - v\|. \blacksquare$$